

《古典概型与古典概型概率计算》习题答案、答案解析与易错点

Exercise Answers, Answer Analysis, and Common Pitfalls for Classical Probability Models

一、选择题（共 20 题）

I. Multiple-Choice Questions (20 questions in total)

第 1 题

题目内容:某人打靶时连续射击两次，下列事件中与事件“至少一次中靶”互为对立的是（ ）

A person shoots twice in a row at a target. Which of the following events is complementary to the event “at least one hit”? ()

A. 至多一次中靶 B. 两次都中靶 C. 只有一次中靶 D. 两次都没有中靶

A. At most one hit B. Two hits C. Exactly one hit D. No hits

答案:D

详细解析: 1. 核心知识点: 对立事件的定义 (两事件必有一个发生且互斥, 即并集为样本空间、交集为空集);

2. 解题步骤: ① 明确“至少一次中靶”的构成: 包含“一次中靶”(第一次中、第二次不中; 第一次不中、第二次中) 和“两次都中靶”两类情况; ② 推导对立事件: 对立事件需包含所有不满足“至少一次中靶”的情况, 即唯一剩余情况“两次都没有中靶”; ③ 样本空间验证: 样本空间 $\Omega = yy, yn, ny, nn$, “至少一次中靶”对应 yy, yn, ny , 其补集为 nn , 即“两次都没有中靶”, 符合对立事件定义;

3. 结论: 故答案为 D。

1. Core Knowledge: Definition of complementary events (two events must occur exactly one and be mutually exclusive, i.e., their union is the sample space and intersection is an empty set);

2. Problem-Solving Steps: ① Clarify the composition of “at least one hit”: including two

categories: "exactly one hit" (hit first, miss second; miss first, hit second) and "two hits";② Derive the complementary event: The complementary event must include all cases that do not satisfy "at least one hit", which is the only remaining case "no hits";③ Sample Space Verification: The sample space $\Omega = yy, yn, ny, nn$, "at least one hit" corresponds to yy, yn, ny , and its complement is nn , i.e., "no hits", which conforms to the definition of complementary events;

3. Conclusion: So the answer is D.

易错点:混淆“对立事件”与“互斥事件”，易错选 A（至多一次中靶）。“至多一次中靶”包含“一次中靶”和“两次都不中靶”，与“至少一次中靶”的交集为“一次中靶”，仅满足互斥（无公共样本点），不满足对立事件“并集为样本空间”的核心条件。

Confusing "complementary events" with "mutually exclusive events" and easily choosing A (at most one hit). "At most one hit" includes "exactly one hit" and "no hits", and its intersection with "at least one hit" is "exactly one hit", which only satisfies mutual exclusivity (no common sample points) but not the core condition of complementary events that "the union is the sample space".

第 2 题

题目内容:从个位数与十位数之和为奇数的两位数中任取一个，其个位数为 0 的概率是（ ）

Take a random two-digit number where the sum of the units digit and the tens digit is odd.

What is the probability that its units digit is 0? ()

A. $\frac{4}{9}$ B. $\frac{1}{3}$ C. $\frac{2}{9}$ D. $\frac{1}{9}$

答案: C

详细解析: 1. 核心知识点：古典概型概率公式（ $P(A) = \frac{\text{事件}A\text{包含的样本点个数}}{\text{样本空间的样本点总数}}$ ）、分类计数原理（加法原理）；

2. 解题步骤：① 确定样本空间（和为奇数的两位数）：两个数之和为奇数的充要条件是“一奇

一偶”，分两类讨论：类别 1：十位数为奇数、个位数为偶数。十位数奇数可选 1,3,5,7,9 (5 种)，个位数偶数可选 0,2,4,6,8 (5 种)，由乘法原理得： $5 \times 5 = 25$ 个；类别 2：十位数为偶数、个位数为奇数。十位数偶数不能为 0，可选 2,4,6,8 (4 种)，个位数奇数可选 1,3,5,7,9 (5 种)，由乘法原理得： $4 \times 5 = 20$ 个；由加法原理，样本空间总数 $= 25 + 20 = 45$ ；

② 确定事件 A (个位数为 0)：个位数为 0 (偶数)，需满足“和为奇数”则十位数必为奇数 (一奇一偶)，十位数奇数有 5 种选择 (1,3,5,7,9)，故事件 A 包含的样本点个数 $= 5$ ；③ 代入古典概型公式计算： $P(A) = \frac{5}{45} = \frac{2}{9}$ ；

3. 结论：故答案为 C。

1. Core Knowledge: Classical probability formula ($P(A) = \frac{\text{Number of sample points in event } A}{\text{Total number of sample points in the sample space}}$), Classification Counting Principle (Addition Principle);

2. Problem-Solving Steps:

① Determine the sample space (two-digit numbers with odd sum): The necessary and sufficient condition for the sum of two numbers to be odd is “one odd and one even”, which is discussed in two categories: Category 1: Tens digit is odd, units digit is even. The odd tens digits can be 1,3,5,7,9 (5 types), and the even units digits can be 0,2,4,6,8 (5 types). By the Multiplication Principle: $5 \times 5 = 25$ numbers; Category 2: Tens digit is even, units digit is odd. The even tens digit cannot be 0, so it can be 2,4,6,8 (4 types), and the odd units digits can be 1,3,5,7,9 (5 types). By the Multiplication Principle: $4 \times 5 = 20$ numbers; By the Addition Principle, the total number of sample points in the sample space $= 25 + 20 = 45$;

② Determine event A (units digit is 0): If the units digit is 0 (even), to satisfy “odd sum”, the tens digit must be odd (one odd and one even). There are 5 choices for the odd tens digit (1,3,5,7,9), so the number of sample points in event A $= 5$;

③ Calculate by substituting into the classical probability formula: $P(A) = \frac{5}{45} = \frac{2}{9}$;

3. Conclusion: So the answer is C.

易错点:1. 遗漏十位数不能为 0 的限制: 计算“十位数为偶数”时误将 0 计入, 导致样本空间总数多算 5 个 (0 为十位时不是两位数), 进而误算概率;

2. 直接枚举计数 (笨算): 未用分类计数原理, 直接罗列所有和为奇数的两位数, 易出现遗漏或重复, 导致样本点个数计算错误。

1. Omitting the restriction that the tens digit cannot be 0: Mistakenly including 0 when calculating “even tens digit”, resulting in 5 more sample points in the sample space (0 as the tens digit is not a two-digit number), and thus miscalculating the probability;

2. Direct enumeration (brute calculation): Not using the Classification Counting Principle, but directly listing all two-digit numbers with odd sum, which is prone to omissions or repetitions, leading to errors in calculating the number of sample points.

第 3 题

题目内容:同时抛掷两枚质地均匀的硬币, 出现两枚都是正面朝上的概率为 ()

Toss two fair coins simultaneously. What is the probability that both land heads up? ()

A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{1}{2}$ D. $\frac{3}{4}$

答案:A

详细解析: 1. 核心知识点: 古典概型 (有限性、等可能性)、样本空间枚举;

2. 解题步骤: ① 确定样本空间: 两枚硬币的所有等可能结果为 $(\text{正}, \text{正}), (\text{正}, \text{反}), (\text{反}, \text{正}), (\text{反}, \text{反})$, 共 4 个样本点; ② 定义事件 A: “两枚都是正面朝上”, 对应样本点 $(\text{正}, \text{正})$, 共 1 个; ③ 代入古典概型公式: $P(A) =$

$$\frac{\text{事件A样本点个数}}{\text{样本空间总数}} = \frac{1}{4};$$

3. 结论: 故答案为 A。

1. Core Knowledge: Classical probability model (finiteness, equiprobability), sample space

enumeration;

2. Problem-Solving Steps: ① Determine the sample space: All equally likely outcomes of two coins are $(Head, Head), (Head, Tail), (Tail, Head), (Tail, Tail)$, totaling 4 sample points;

② Define event A: "Both land heads up", corresponding to the sample point $(Head, Head)$, totaling 1;

③ Substitute into the classical probability formula: $P(A) = \frac{\text{Number of sample points in event } A}{\text{Total number of sample points in the sample space}} = \frac{1}{4}$; 3.

Conclusion: So the answer is A.

易错点:误将样本空间简化为 全正, 全反, 一正一反 (3个样本点), 忽略“一正一反”包含(正, 反)和(反, 正)两种等可能情况, 导致误算概率为 $\frac{1}{3}$ 。

Mistakenly simplifying the sample space to $BothHeads, BothTails, OneHeadOneTail$ (3 sample points), ignoring that "One Head One Tail" includes two equally likely cases: (Head, Tail) and (Tail, Head), leading to a miscalculated probability of $\frac{1}{3}$.

第4题

题目内容: 国庆阅兵中, 某兵种 A, B, C 三个方阵按随机次序通过主席台, 则 B 先于 A, C 通过的概率为 ()

In the National Day military parade, three phalanxes A, B, and C of a certain arm pass the rostrum in random order. What is the probability that B passes before A and C? ()

A. $\frac{1}{6}$ B. $\frac{1}{3}$ C. $\frac{1}{2}$ D. $\frac{2}{3}$

答案:B

详细解析: 1. 核心知识点: 古典概型、对称法 (排列的等可能性);

2. 解题步骤: ① 确定样本空间: 三个方阵的全排列数为 $3! = 6$, 所有排列为 $ABC, ACB, BAC, BCA, CAB, CBA$; ② 利用对称法分析: 在所有排列中, “B先于A和C”“A先于B和C”“C先于A和B”这三种情况是等可能的 (无任何偏向性); ③ 计算概率: 三种等可能情

况各占 $\frac{1}{3}$ ，故“B 先于 A 和 C”的概率为 $\frac{1}{3}$ ；

3. 结论：故答案为 B。

1. Core Knowledge: Classical probability model, symmetry method (equipotentiality of permutations);

2. Problem-Solving Steps: ① Determine the sample space: The total number of permutations of three phalanxes is $3! = 6$ ，and all permutations are $ABC, ACB, BAC, BCA, CAB, CBA$ ；②

Analyze with symmetry: Among all permutations, the three cases “B before A and C”, “A before B and C”, and “C before A and B” are equally likely (no bias); ③ Calculate the probability: Each

of the three equally likely cases accounts for $\frac{1}{3}$ ，so the probability that “B passes before A and C” is $\frac{1}{3}$ ；

3. Conclusion: So the answer is B.

易错点:1. 枚举排列时遗漏或重复（如漏算 BCA 或误加重重复排列）；2. 未利用对称法，逐一枚举后误将符合条件的排列数算为 1（如仅识别 BAC，漏算 BCA），导致概率误算为 $\frac{1}{6}$ 。

1. Omitting or repeating permutations during enumeration (e.g., missing BCA or adding duplicate permutations); 2. Not using the symmetry method and mistakenly counting only 1 valid permutation (e.g., recognizing only BAC and missing BCA) after enumeration, leading to a miscalculated probability of $\frac{1}{6}$.

第 5 题

题目内容:甲、乙两位同学考入同一专业（设 3 个班级），他们被随机分到同一个班级的概率为

Two students, A and B, are admitted to the same major (with 3 classes). What is the probability that they are randomly assigned to the same class? ()

A. $\frac{1}{9}$ B. $\frac{1}{6}$ C. $\frac{1}{3}$ D. $\frac{1}{2}$

答案:C

详细解析: 1. 核心知识点: 古典概型、分步计数原理;

2. 解题步骤: ① 确定样本空间: 甲有 3 种班级选择, 乙独立也有 3 种选择, 总样本点个数为 $3 \times 3 =$

9 ; ② 定义事件 A: “分到同一个班级”, 对应样本点为 (1 班, 1 班), (2 班, 2 班), (3 班, 3 班) , 共

3 个; ③ 代入公式: $P(A) = \frac{3}{9} = \frac{1}{3}$;

3. 结论: 故答案为 C。

1. Core Knowledge: Classical probability model, step-by-step counting principle;

2. Problem-Solving Steps: ① Determine the sample space: Student A has 3 class choices, and

Student B also has 3 independent choices, so the total number of sample points is $3 \times 3 =$

9 ; ② Define event A: “Assigned to the same class”, corresponding to sample points

(Class1, Class1), (Class2, Class2), (Class3, Class3) , totaling 3; ③ Substitute into the formula:

$$P(A) = \frac{3}{9} = \frac{1}{3} ;$$

3. Conclusion: So the answer is C.

易错点: 误将样本空间视为“从 3 个班级选 2 个分配” (组合问题), 计算样本点个数为 $C_3^2 \times 2 =$

6 , 忽略甲、乙选择的独立性 (允许同班级), 导致概率误算为 $\frac{3}{6} = \frac{1}{2}$ 。

Mistakenly regarding the sample space as “selecting 2 classes from 3 for assignment”

(combination problem), calculating the total number of sample points as $C_3^2 \times 2 = 6$, and

ignoring the independence of A and B's choices (allowing the same class), leading to a

miscalculated probability of $\frac{3}{6} = \frac{1}{2}$.

第 6 题

题目内容: 从 20 名男同学、10 名女同学中任选 3 名参加体能测试, 选到的 3 名同学中既有男

同学又有女同学的概率为 ()

Select 3 students randomly from 20 male students and 10 female students to participate in a physical fitness test. What is the probability that the selected 3 students include both male and female students? ()

- A. $\frac{20}{29}$ B. $\frac{19}{29}$ C. $\frac{10}{29}$ D. $\frac{9}{29}$

答案:A

详细解析: 1. 核心知识点: 古典概型、对立事件 (简化计算);

2. 解题步骤: ① 确定对立事件: “既有男又有女”的对立事件是“全为男生”或“全为女生”; ② 计

算总样本数: 从 30 名学生选 3 名, 组合数 $C_{30}^3 = \frac{30 \times 29 \times 28}{3 \times 2 \times 1} = 4060$; ③ 计算对立事件样本数:

全男生 $C_{20}^3 = 1140$, 全女生 $C_{10}^3 = 120$, 合计 $1140 + 120 = 1260$; ④ 代入公式:

$$P(\text{既有男又有女}) = 1 - P(\text{全男或全女}) = 1 - \frac{1260}{4060} = 1 - \frac{9}{29} = \frac{20}{29};$$

3. 结论: 故答案为 A。

1. Core Knowledge: Classical probability model, complementary events (simplified calculation);

2. Problem-Solving Steps: ① Determine the complementary event: The complementary event

of “both male and female” is “all male” or “all female”; ② Calculate the total number of sample

points: The combination number of selecting 3 from 30 students is $C_{30}^3 = \frac{30 \times 29 \times 28}{3 \times 2 \times 1} = 4060$; ③

Calculate the number of sample points for the complementary event: All male $C_{20}^3 = 1140$, all

female $C_{10}^3 = 120$, totaling $1140 + 120 = 1260$; ④ Substitute into the formula:

$$P(\text{Both male and female}) = 1 - P(\text{All male or all female}) = 1 - \frac{1260}{4060} = 1 - \frac{9}{29} = \frac{20}{29}; \quad 3.$$

Conclusion: So the answer is A.

易错点: 1. 直接计算“1 男 2 女”和“2 男 1 女”的概率时, 组合数计算错误 (如 $C_{20}^1 \times C_{10}^2$ 算错);

2. 混淆“组合”与“排列”, 用排列数计算样本数 (如分母用 $30 \times 29 \times 28$), 导致结果错误。

1. Miscalculating combination numbers when directly calculating the probability of “1 male and 2 females” and “2 males and 1 female” (e.g., miscalculating $C_{20}^1 \times C_{10}^2$); 2. Confusing “combination” with “permutation” and using permutation numbers to calculate sample points (e.g., using $30 \times 29 \times 28$ as the denominator), leading to incorrect results.

第 7 题

题目内容:有 5 本不同的书 (2 本语文、2 本数学、1 本物理), 随机并排摆放在书架上, 同一科目的书都不相邻的概率为 ()

There are 5 different books (2 Chinese, 2 Math, and 1 Physics). They are randomly placed side by side on a bookshelf. What is the probability that books of the same subject are not adjacent? ()

A. $\frac{1}{5}$ B. $\frac{2}{5}$ C. $\frac{3}{5}$ D. $\frac{4}{5}$

答案:B

详细解析: 1. 核心知识点: 古典概型、容斥原理 (计算对立事件);

2. 解题步骤: ① 总样本数: 5 本不同书的全排列数 $5! = 120$; ② 定义对立事件: “至少有一类科目书相邻”, 设 A = “语文书相邻”, B = “数学书相邻”;

③ 用容斥原理计算 $P(A \cup B)$: $P(A)$: 将 2 本语文书视为整体, 4 个元素排列数 $4! \times 2! = 48$, 故 $P(A) = \frac{48}{120}$; $P(B)$: 同理, $P(B) = \frac{4! \times 2!}{120} = \frac{48}{120}$; $P(A \cap B)$: 语文、数学书均视为整体, 3 个元素排列数 $3! \times 2! \times 2! = 24$, 故 $P(A \cap B) = \frac{24}{120}$; 由容斥原理:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{48+48-24}{120} = \frac{72}{120};$$

④ 计算目标概率: $P(\text{同科目都不相邻}) = 1 - \frac{72}{120} = \frac{48}{120} = \frac{2}{5}$; 3. 结论: 故答案为 B。

1. Core Knowledge: Classical probability model, inclusion-exclusion principle (calculating complementary events); 2. Problem-Solving Steps: ① Total number of sample points: The total permutation number of 5 different books is $5! = 120$; ② Define the complementary event: “At least one subject’s books are adjacent”, let A = “Chinese books are adjacent”, B = “Math

books are adjacent"; ③ Calculate $P(A \cup B)$ using the inclusion-exclusion principle: $P(A)$: Treat

2 Chinese books as a whole, the number of permutations of 4 elements is $4! \times 2! = 48$, so

$P(A) = \frac{48}{120}$; $P(B)$: Similarly, $P(B) = \frac{4! \times 2!}{120} = \frac{48}{120}$; $P(A \cap B)$: Treat both Chinese and Math

books as wholes, the number of permutations of 3 elements is $3! \times 2! \times 2! = 24$, so

$P(A \cap B) = \frac{24}{120}$; By inclusion-exclusion principle: $P(A \cup B) = P(A) + P(B) - P(A \cap B) =$

$$\frac{48+48-24}{120} = \frac{72}{120};$$

④ Calculate the target probability: $P(\text{No same subject adjacent}) = 1 - \frac{72}{120} = \frac{48}{120} = \frac{2}{5}$; 3.

Conclusion: So the answer is B.

易错点:1. 直接用插空法时考虑不全 (如先排物理书, 再插语文和数学书, 忽略数学书可能相邻); 2. 容斥原理中重复计算 $P(A \cap B)$ (如误将 $A \cap B$ 的排列数算为 $3! \times 2! = 12$, 漏乘数学书的排列数 $2!$)。

1. Incomplete consideration when directly using the gap method (e.g., arranging Physics books first and then inserting Chinese and Math books, ignoring that Math books may be adjacent); 2.

Double-counting $P(A \cap B)$ in the inclusion-exclusion principle (e.g., mistakenly calculating the permutation number of $A \cap B$ as $3! \times 2! = 12$, missing the multiplication by the permutation number of Math books $2!$).

第 8 题

题目内容:一个盒子有 6 个小格子, 现有桔子、苹果、香蕉各 2 个, 将其随机放入格子 (每个格子 1 个), 放好后每行、每列水果种类各不相同的概率是 ()

A box has 6 small grids. There are 2 oranges, 2 apples, and 2 bananas. They are randomly placed into the grids (1 per grid). What is the probability that the types of fruits in each row and column are different after placement? ()

A. $\frac{2}{15}$ B. $\frac{2}{9}$ C. $\frac{1}{5}$ D. $\frac{1}{3}$

答案:A

详细解析: 1. 核心知识点: 古典概型、分步计数 (基于 2 行 3 列格子结构, 常见题型默

认); 2. 解题步骤: ① 总样本数: 6 个水果的全排列数 $6! = 720$ (水果视为不同个体); ②

事件 A (每行每列种类不同) 的分步计算: 第一步: 排第 1 行 (3 个格子), 需 3 种水果各 1

个, 排列数 $3! = 6$, 从每种水果的 2 个中选 1 个, 选择数 $2 \times 2 \times 2 = 8$, 共 $6 \times 8 = 48$

种; 第二步: 排第 2 行 (3 个格子), 需满足“每列与第 1 行种类不同”, 即 3 个元素的错位排

列 (derangement), 错位排列数为 2 (如第 1 行是{桔, 苹, 香}, 第 2 行只能是{苹, 香, 桔}或{香,

桔, 苹}), 剩余水果各 1 个, 仅 1 种选择, 共 $2 \times 1 = 2$ 种; 事件 A 样本数: $48 \times 2 = 96$;

③ 计算概率: $P(A) = \frac{96}{720} = \frac{2}{15}$;

3. 结论: 故答案为 A。

1. Core Knowledge: Classical probability model, step-by-step counting (based on 2-row 3-

column grid structure, default for common questions); 2. Problem-Solving Steps: ① Total

number of sample points: The total permutation number of 6 fruits is $6! = 720$ (fruits are

regarded as different individuals); ② Step-by-step calculation of event A (different types in each

row and column): Step 1: Arrange the first row (3 grids), requiring 1 fruit of each type, with

permutation number $3! = 6$. Select 1 from 2 of each fruit, with selection number $2 \times 2 \times 2 =$

8, totaling $6 \times 8 = 48$ ways; Step 2: Arrange the second row (3 grids), requiring “different types

from the first row in each column”, i.e., derangement of 3 elements. The derangement number

is 2 (e.g., if the first row is {Orange, Apple, Banana}, the second row can only be {Apple, Banana,

Orange} or {Banana, Orange, Apple}). Only 1 choice for the remaining 1 fruit of each type,

totaling $2 \times 1 = 2$ ways; Number of sample points for event A: $48 \times 2 = 96$; ③ Calculate the

probability: $P(A) = \frac{96}{720} = \frac{2}{15}$;

3. Conclusion: So the answer is A.

易错点:1. 格子结构不明确时无法计算 (需默认 2 行 3 列, 否则逻辑不通); 2. 错位排列数算错 (3 个元素错位排列数为 2, 非 1 或 3); 3. 忽略水果的选择性 (误将水果视为相同, 总样本数算为 $\frac{6!}{2! \times 2! \times 2!} = 90$, 导致概率误算为 $\frac{12}{90} = \frac{2}{15}$ (结果巧合但逻辑错误))。

1. Unable to calculate without a clear grid structure (2-row 3-column is default, otherwise illogical); 2. Miscalculating the derangement number (the derangement number of 3 elements is 2, not 1 or 3); 3. Ignoring fruit selectivity (mistakenly regarding fruits as identical, calculating the total number of sample points as $\frac{6!}{2! \times 2! \times 2!} = 90$, leading to a coincidentally correct result of $\frac{12}{90} = \frac{2}{15}$ but incorrect logic).

第 9 题

题目内容:铸币大臣在每箱 100 枚硬币中掺入 1 枚劣币, 国王用两种方法检测: 方法一 (10 箱各抽 1 枚)、方法二 (5 箱各抽 2 枚), 能发现至少 1 枚劣币的概率分别为 p_1, p_2 , 则 ()

The Minister of Coinage mixes 1 counterfeit coin into each box of 100 coins. The king uses two detection methods: Method 1 (draw 1 coin from each of 10 boxes) and Method 2 (draw 2 coins from each of 5 boxes). The probabilities of finding at least 1 counterfeit coin are p_1 and p_2 respectively. Then ()

A. $p_1 = p_2$ B. $p_1 < p_2$ C. $p_1 > p_2$ D. 无法确定

答案:B

详细解析: 1. 核心知识点: 古典概型、对立事件 (至少 1 枚劣币=1-全为真币); 2. 解题步

骤: ① 计算 p_1 (方法一): 每箱抽 1 枚全为真币的概率: $\left(\frac{99}{100}\right)^{10}$; $p_1 = 1 - \left(\frac{99}{100}\right)^{10}$;

② 计算 p_2 (方法二): 每箱抽 2 枚全为真币的概率: $\frac{C_{99}^2}{C_{100}^2} = \frac{99 \times 98}{100 \times 99} = \frac{49}{50}$; 5 箱全为真币的概率: $\left(\frac{49}{50}\right)^5$; $p_2 = 1 - \left(\frac{49}{50}\right)^5$;

③ 比较 p_1 和 p_2 : 变形: $\left(\frac{99}{100}\right)^{10} = \left(\frac{9801}{10000}\right)^5$, $\left(\frac{49}{50}\right)^5 = \left(\frac{9800}{10000}\right)^5$; 因 $\frac{9801}{10000} > \frac{9800}{10000}$, 故 $\left(\frac{9801}{10000}\right)^5 > \left(\frac{9800}{10000}\right)^5$; 进而 $1 - \text{前者} < 1 - \text{后者}$, 即 $p_1 < p_2$;

3. 结论: 故答案为 B.

1. Core Knowledge: Classical probability model, complementary events (at least 1 counterfeit = 1 - all genuine);

2. Problem-Solving Steps: ① Calculate p_1 (Method 1): Probability of drawing all genuine coins (1 from each of 10 boxes): $\left(\frac{99}{100}\right)^{10}$; $p_1 = 1 - \left(\frac{99}{100}\right)^{10}$;

② Calculate p_2 (Method 2): Probability of drawing all genuine coins (2 from one box): $\frac{C_{99}^2}{C_{100}^2} = \frac{99 \times 98}{100 \times 99} = \frac{49}{50}$; Probability of drawing all genuine coins from 5 boxes: $\left(\frac{49}{50}\right)^5$; $p_2 = 1 - \left(\frac{49}{50}\right)^5$;

③ Compare p_1 and p_2 : Transform: $\left(\frac{99}{100}\right)^{10} = \left(\frac{9801}{10000}\right)^5$, $\left(\frac{49}{50}\right)^5 = \left(\frac{9800}{10000}\right)^5$; Since $\frac{9801}{10000} > \frac{9800}{10000}$, then $\left(\frac{9801}{10000}\right)^5 > \left(\frac{9800}{10000}\right)^5$; Further, $1 - \text{the former} < 1 - \text{the latter}$, i.e., $p_1 < p_2$;

3. Conclusion: So the answer is B.

易错点: 1. 直接计算“至少 1 枚劣币”的概率 (如方法一用 $C_{10}^1 \times \frac{1}{100} \times \left(\frac{99}{100}\right)^9 + \dots$), 计算复杂且易出错; 2. 幂运算比较时误判 $\left(\frac{99}{100}\right)^{10}$ 与 $\left(\frac{49}{50}\right)^5$ 的大小 (如误以为两者相等)。

1. Directly calculating the probability of “at least 1 counterfeit” (e.g., Method 1 using $C_{10}^1 \times \frac{1}{100} \times \left(\frac{99}{100}\right)^9 + \dots$), which is complex and error-prone; 2. Misjudging the size of $\left(\frac{99}{100}\right)^{10}$ and $\left(\frac{49}{50}\right)^5$ during power comparison (e.g., mistakenly thinking they are equal).

第 10 题

题目内容:从三男三女 6 名学生中任选 2 名 (每名被选中机会相等), 则 2 名都是女同学的概率等于 ()

Select 2 students randomly from 6 students (3 boys and 3 girls) with equal chance of being selected. What is the probability that both selected students are female? ()

A. $\frac{1}{5}$ B. $\frac{1}{3}$ C. $\frac{1}{2}$ D. $\frac{2}{3}$

答案:A

详细解析: 1. 核心知识点: 古典概型、组合计数;

2. 解题步骤: ① 总样本数: 从 6 名学生选 2 名, 组合数 $C_6^2 = \frac{6 \times 5}{2 \times 1} = 15$; ② 事件 A (2 名都是女同学) 的样本数: 从 3 名女生选 2 名, 组合数 $C_3^2 = 3$; ③ 代入公式: $P(A) = \frac{3}{15} = \frac{1}{5}$;

3. 结论: 故**答案**为 A。

1. Core Knowledge: Classical probability model, combination counting;

2. Problem-Solving Steps: ① Total number of sample points: The combination number of selecting 2 from 6 students is $C_6^2 = \frac{6 \times 5}{2 \times 1} = 15$; ② Number of sample points for event A (both female): The combination number of selecting 2 from 3 girls is $C_3^2 = 3$; ③ Substitute into the formula: $P(A) = \frac{3}{15} = \frac{1}{5}$;

3. Conclusion: So the answer is A.

易错点:1. 用排列数计算样本数 (如分母用 $6 \times 5 = 30$, 分子用 $3 \times 2 = 6$), 虽结果 $\frac{6}{30} = \frac{1}{5}$ 正确, 但逻辑错误 (选 2 名学生无顺序); 2. 误将样本数算为 $6 + 5 = 11$ (加法计数错误), 导致概率误算。

1. Using permutation numbers to calculate sample points (e.g., denominator $6 \times 5 = 30$,

numerator $3 \times 2 = 6$), resulting in a coincidentally correct result of $\frac{6}{30} = \frac{1}{5}$ but incorrect logic (no order in selecting 2 students); 2. Mistakenly calculating the number of sample points as $6 + 5 = 11$ (addition counting error), leading to miscalculated probability.

第 11 题

题目内容:从 n 个正整数 $1, 2, \dots, n$ 中任意取出两个不同的数, 若两数之和等于 5 的概率为 $\frac{1}{14}$, 则 $n = ()$

Take two different numbers randomly from n positive integers $1, 2, \dots, n$. If the probability that the sum of the two numbers is 5 is $\frac{1}{14}$, then $n = ()$

A. 6 B. 7 C. 8 D. 9

答案:C

C

详细解析:

1. 核心知识点: 古典概型、组合计数;

2. 解题步骤: ① 确定和为 5 的数对: 正整数中两数之和为 5 的不同数对仅 $(1, 4), (2, 3)$, 共

2 个; ② 总样本数: 从 n 个数中取 2 个的组合数为 $C_n^2 = \frac{n(n-1)}{2}$; ③ 列概率方程: 由古典

概型公式, $\frac{2}{C_n^2} = \frac{1}{14}$, 代入组合数公式得 $\frac{2}{\frac{n(n-1)}{2}} = \frac{1}{14}$; ④ 解方程: 化简得 $n(n-1) = 56$,

解得 $n = 8$ ($n = -7$ 舍去);

3. 结论: 故**答案**为 C。

1. Core Knowledge: Classical probability model, combination counting;

2. Problem-Solving Steps: ① Determine number pairs with sum 5: The only distinct pairs of positive integers summing to 5 are $(1, 4), (2, 3)$, totaling 2; ② Total number of sample points:

The combination number of selecting 2 from n numbers is $C_n^2 = \frac{n(n-1)}{2}$; ③ Set up probability equation: By the classical probability formula, $\frac{2}{C_n^2} = \frac{1}{14}$. Substitute the combination formula: $\frac{2}{\frac{n(n-1)}{2}} = \frac{1}{14}$; ④ Solve the equation: Simplify to $n(n-1) = 56$, solve for $n = 8$ ($n = -7$ is discarded);

3. Conclusion: So the answer is C.

易错点:1. 漏算和为 5 的数对 (如仅识别 (1,4), 漏 (2,3)), 导致方程左边分子为 1, 误解得 $n = 5$; 2. 组合数公式记忆错误 (如写成 $C_n^2 = n(n-1)$), 导致解方程出错。

1. Missing number pairs with sum 5 (e.g., recognizing only (1,4) and missing (2,3)), resulting in the numerator on the left side of the equation being 1 and miscalculating $n = 5$; 2. Misremembering the combination formula (e.g., writing $C_n^2 = n(n-1)$), leading to errors in solving the equation.

第 12 题

题目内容:将 3 个相同黑球和 3 个相同白球自左向右排成一排, 若“从任意位置向左数, 黑球个数 $\geq$ 白球个数”为有效排列, 则有效排列的概率为 ()

Arrange 3 identical black balls and 3 identical white balls in a row from left to right. A “valid arrangement” is defined as “from any position, the number of black balls counted to the left is $\geq$ the number of white balls”. What is the probability of a valid arrangement? ()

A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{1}{5}$ D. $\frac{1}{10}$

答案:C

详细解析:

1. 核心知识点: 古典概型、卡特兰数 (有效排列计数);

2. 解题步骤: ① 总样本数: 6 个位置中选 3 个放黑球 (白球位置固定), 组合数 $C_6^3 = 20$;

② 有效排列数：此类“任意前缀中黑球 $\geq$ 白球”的排列为卡特兰数问题，当黑球数=白球数= k

时，有效排列数为卡特兰数 $C_k = \frac{1}{k+1} C_{2k}^k$ ； ③ 计算卡特兰数： $k = 3$ 时， $C_3 = \frac{1}{4} C_6^3 =$

$\frac{1}{4} \times 20 = 5$ ； ④ 计算概率： $P = \frac{5}{20} = \frac{1}{5}$ ；

3. 结论：故答案为 C。

1. Core Knowledge: Classical probability model, Catalan number (counting valid arrangements);

2. Problem-Solving Steps: ① Total number of sample points: Select 3 positions out of 6 for black balls (white ball positions are fixed), combination number $C_6^3 = 20$ ； ② Number of valid

arrangements: Such arrangements where “black balls $\geq$ white balls in any prefix” are a Catalan

number problem. When the number of black balls = number of white balls = k , the number

of valid arrangements is the Catalan number $C_k = \frac{1}{k+1} C_{2k}^k$ ； ③ Calculate Catalan number: For

$k = 3$, $C_3 = \frac{1}{4} C_6^3 = \frac{1}{4} \times 20 = 5$ ； ④ Calculate probability: $P = \frac{5}{20} = \frac{1}{5}$ ；

3. Conclusion: So the answer is C.

易错点:1. 未掌握卡特兰数，直接枚举有效排列时遗漏（如漏算“黑、黑、白、黑、白、白”）；

2. 误将球视为不同个体，总样本数算为 $6! = 720$ ，导致概率计算错误。

1. Not mastering Catalan numbers and missing valid arrangements when enumerating directly

(e.g., omitting “Black, Black, White, Black, White, White”); 2. Mistakenly regarding balls as distinct

individuals and calculating the total number of sample points as $6! = 720$, leading to incorrect

probability calculation.

第 13 题

题目内容:已知点 $P_1, P_2, \dots, P_{10}$ 分别是四面体的顶点或棱的中点，四点组 (P_1, P_i, P_j, P_k) ($1 <$

$i < j < k \leq 10$) 在同一平面上的概率为 ()

Given that points $P_1, P_2, \dots, P_{10}$ are respectively the vertices or midpoints of the edges of a

tetrahedron. What is the probability that the four-point group (P_1, P_i, P_j, P_k) ($1 < i < j < k \leq 10$) lies on the same plane? ()

- A. $\frac{2}{5}$ B. $\frac{3}{7}$ C. $\frac{4}{9}$ D. $\frac{5}{11}$

答案:B

详细解析:

1. 核心知识点: 古典概型、四面体的结构性质;

2. 解题步骤: ① 总样本数: 固定 P_1 , 从剩余 9 个点中选 3 个, 组合数 $C_9^3 = 84$; ② 共面

四点组计数 (含 P_1): 类别 1: P_1 所在的四面体的面: 四面体有 4 个面, 每个面含 6 个点

(4 个顶点+2 条棱中点), 从每个面剩余 5 个点选 2 个, 共 $4 \times C_5^2 = 40$; 类别 2: P_1 所在

棱与对棱中点构成的面: 四面体有 3 条棱含 P_1 , 每条棱的对棱中点与该棱构成 1 个面, 每个

面含 3 个额外点, 选 2 个, 共 $3 \times C_3^2 = 9$; 总共面样本数: $40 + 9 = 49$; ③ 计算概率:

$$P = \frac{49}{84} = \frac{3}{7};$$

3. 结论: 故答案为 B。

1. Core Knowledge: Classical probability model, structural properties of tetrahedrons;

2. Problem-Solving Steps: ① Total number of sample points: Fix P_1 and select 3 from the remaining 9 points, combination number $C_9^3 = 84$; ② Count coplanar four-point groups

(including P_1): Category 1: Faces of the tetrahedron containing P_1 : A tetrahedron has 4 faces, each containing 6 points (4 vertices + 2 edge midpoints). Select 2 from the remaining 5 points

on each face, totaling $4 \times C_5^2 = 40$; Category 2: Faces formed by the edge containing P_1 and

the midpoint of its opposite edge: A tetrahedron has 3 edges containing P_1 , each edge forms

1 face with the midpoint of its opposite edge, and each face contains 3 additional points. Select

2, totaling $3 \times C_3^2 = 9$; Total coplanar sample points: $40 + 9 = 49$; ③ Calculate probability:

$$P = \frac{49}{84} = \frac{7}{12} ;$$

3. Conclusion: So the answer is B.

易错点:1. 漏算“棱与对棱中点构成的面”（类别 2），导致共面样本数偏少，概率误算；2. 误将四面体的面含有的点数算为 4（仅顶点），忽略棱中点，导致计数错误。

1. Missing “faces formed by edges and midpoints of opposite edges” (Category 2), resulting in fewer coplanar sample points and miscalculated probability; 2. Mistakenly calculating the number of points on a tetrahedron's face as 4 (only vertices) and ignoring edge midpoints, leading to counting errors.

第 14 题

题目内容:东西方向有 4 条道路, 南北方向有 5 条道路, 从点 A 出发沿最短道路行走至 C 点, 中途不经过点 P 的概率是 ()

There are 4 roads in the east-west direction and 5 roads in the north-south direction. Starting from point A , walk along the shortest path to point C . What is the probability of not passing through point P midway? ()

A. $\frac{3}{7}$ B. $\frac{4}{9}$ C. $\frac{5}{11}$ D. $\frac{6}{13}$

答案:A

详细解析:

- 核心知识点：古典概型、组合计数（最短路径核心原理）；
- 解题步骤：① 最短路径原理：从 A 到 C 的最短路径，需向东走 3 段（4 条东西道路间隔 3 段）、向北走 4 段（5 条南北道路间隔 4 段），总路径数为组合数 $C_{3+4}^3 = C_7^3 = 35$ ；② 计算经过点 P 的路径数：最短路径中经过 P 可拆分为 $A \rightarrow P$ 和 $P \rightarrow C$ 两段独立路径， $A \rightarrow P$ 需向东 1 段、向北 2 段，路径数 $C_3^1 = 3$ ； $P \rightarrow C$ 需向东 2 段、向北 2 段，路径数 $C_4^2 = 6$ ，

故经过 P 的路径总数为 $3 \times 6 = 20$; ③ 计算不经过 P 的路径数: $35 - 20 = 15$; ④ 代

入公式求概率: $P = \frac{15}{35} = \frac{3}{7}$;

3. 结论: 故答案为 A。

1. Core Knowledge: Classical probability model, combination counting (core principle of shortest path);

2. Problem-Solving Steps: ① Shortest path principle: The shortest path from A to C requires walking 3 east segments (3 gaps between 4 east-west roads) and 4 north segments (4 gaps between 5 north-south roads). The total number of paths is the combination number $C_{3+4}^3 = C_7^3 = 35$; ② Calculate the number of paths passing through P : The shortest path passing through P can be split into two independent paths $A \rightarrow P$ and $P \rightarrow C$. $A \rightarrow P$ needs 1 east segment and 2 north segments with path number $C_3^1 = 3$; $P \rightarrow C$ needs 2 east segments and 2 north segments with path number $C_4^2 = 6$. The total number of paths passing through P is $3 \times 6 = 20$; ③ Calculate the number of paths not passing through P : $35 - 20 = 15$; ④ Calculate probability by formula: $P = \frac{15}{35} = \frac{3}{7}$;

3. Conclusion: So the answer is A.

易错点:1. 最短路径步数计算错误 (如将向东步数算为 4); 2. 拆分 $A \rightarrow P$ 和 $P \rightarrow C$ 时, 方向步数对应错误, 导致路径数计数偏差; 3. 未用对立事件思想, 直接枚举不经过 P 的路径, 易遗漏。

1. Miscalculating the number of steps for the shortest path (e.g., counting east steps as 4); 2. Incorrectly matching direction steps when splitting $A \rightarrow P$ and $P \rightarrow C$, leading to deviations in path counting; 3. Not using the complementary event idea and directly enumerating paths not passing through P , which is prone to omissions.

第 15 题

题目内容:将号码 1~9 的九个小球放入袋中, 甲摸出号码 a 放回后, 乙摸出号码 b , 则不等式 $a - 2b + 10 > 0$ 成立的概率为 ()

Put nine small balls numbered 1~9 into a bag. After A draws a ball with number a and puts it back, B draws a ball with number b . What is the probability that the inequality $a - 2b + 10 > 0$ holds? ()

- A. $\frac{52}{81}$ B. $\frac{59}{81}$ C. $\frac{60}{81}$ D. $\frac{61}{81}$

答案:D

详细解析:

1. 核心知识点: 古典概型、分步计数 (有放回抽样); 2. 解题步骤: ① 总样本数: 有放回抽样, 甲、乙各有 9 种选择, 总样本点个数为 $9 \times 9 = 81$; ② 不等式变形: $a - 2b + 10 > 0 \Rightarrow a > 2b - 10$, 其中 $a, b \in 1, 2, \dots, 9$; ③ 按 b 的取值分类, 统计满足条件的 a 的个数:

- $b = 1$: $a > -8 \rightarrow a = 1 - 9$, 共 9 个;
- $b = 2$: $a > -6 \rightarrow a = 1 - 9$, 共 9 个;
- $b = 3$: $a > -4 \rightarrow a = 1 - 9$, 共 9 个;
- $b = 4$: $a > -2 \rightarrow a = 1 - 9$, 共 9 个;
- $b = 5$: $a > 0 \rightarrow a = 1 - 9$, 共 9 个;
- $b = 6$: $a > 2 \rightarrow a = 3 - 9$, 共 7 个;
- $b = 7$: $a > 4 \rightarrow a = 5 - 9$, 共 5 个;
- $b = 8$: $a > 6 \rightarrow a = 7 - 9$, 共 3 个;
- $b = 9$: $a > 8 \rightarrow a = 9$, 共 1 个; 满足条件的总个数: $9 \times 5 + 7 + 5 + 3 + 1 = 45 + 16 = 61$;

④ 代入公式计算概率: $P = \frac{61}{81}$; 3. 结论: 故答案为 D。

1. Core Knowledge: Classical probability model, step-by-step counting (with-replacement sampling); 2. Problem-Solving Steps: ① Total number of sample points: With-replacement sampling, A and B each have 9 choices, total sample points $9 \times 9 = 81$; ② Inequality transformation: $a - 2b + 10 > 0 \Rightarrow a > 2b - 10$, where a and b are integers from 1 to 9; ③

Count valid a for each b :

- $b = 1$: $a > -8 \rightarrow a = 1 \sim 9$ (9 values);
- $b = 2$: $a > -6 \rightarrow a = 1 \sim 9$ (9 values);
- $b = 3$: $a > -4 \rightarrow a = 1 \sim 9$ (9 values);
- $b = 4$: $a > -2 \rightarrow a = 1 \sim 9$ (9 values);
- $b = 5$: $a > 0 \rightarrow a = 1 \sim 9$ (9 values);
- $b = 6$: $a > 2 \rightarrow a = 3 \sim 9$ (7 values);
- $b = 7$: $a > 4 \rightarrow a = 5 \sim 9$ (5 values);
- $b = 8$: $a > 6 \rightarrow a = 7 \sim 9$ (3 values);
- $b = 9$: $a > 8 \rightarrow a = 9$ (1 value); Total valid a : $9 \times 5 + 7 + 5 + 3 + 1 = 45 + 16 = 61$;

④ Calculate probability: $P = \frac{61}{81}$;

3. Conclusion: So the answer is D.

易错点: 1. 不等式变形错误 (如写成 $a < 2b - 10$), 导致计数完全相反; 2. 漏算 $b = 1 \sim 5$ 时 a 的全部取值 (如误将 $b = 5$ 时 $a > 0$ 算为 8 个); 3. 重复计数 a 的取值, 导致有效个数偏多。

1. Incorrect inequality transformation (e.g., writing $a < 2b - 10$), resulting in completely

opposite counting; 2. Missing all values of a when $b = 1 \sim 5$ (e.g., mistakenly counting $a > 0$ as 8 values when $b = 5$); 3. Double-counting values of a , leading to more valid counts than actual.

第 16 题

题目内容:一个口袋有 5 个小球 (2 红 2 白 1 黑), 依次摸出 5 个球, 相邻两个小球颜色均不同的概率是 ()

A pocket has 5 small balls (2 red, 2 white, 1 black). Draw 5 balls in sequence. What is the probability that adjacent balls have different colors? ()

A. $\frac{2}{5}$ B. $\frac{1}{5}$ C. $\frac{3}{5}$ D. $\frac{4}{5}$

答案:B

详细解析:

1. 核心知识点: 古典概型、相同元素排列;

2. 解题步骤: ① 总样本数: 5 个球 (2 红 2 白 1 黑) 的全排列数 (相同元素) 为 $\frac{5!}{2!2!1!} = 30$;

② 有效排列条件: 相邻颜色不同, 关键在黑球位置 (仅 1 个, 可分隔红白); ③ 分黑球位置计数:

- 黑球在位置 2: 剩余位置 1、3、4、5, 需排 2 红 2 白且相邻不同, 仅 1 种排列 (红、黑、白、红、白);
- 黑球在位置 3: 剩余位置 1、2、4、5, 排 2 红 2 白且相邻不同, 有 2 种排列 (红、白、黑、红、白; 白、红、黑、白、红);
- 黑球在位置 4: 同位置 2, 仅 1 种排列 (红、白、红、黑、白);
- 黑球在位置 1: 排列为 (黑、红、白、红、白), 有效;
- 黑球在位置 5: 排列为 (红、白、红、白、黑), 有效; 有效排列数: $1+2+1+1+1=6$;

④ 计算概率: $P = \frac{6}{30} = \frac{1}{5}$; 3. 结论: 故答案为 B。

1. Core Knowledge: Classical probability model, permutation of identical elements; 2. Problem-

Solving Steps: ① Total number of sample points: The total permutation number of 5 balls (2 red,

2 white, 1 black) with identical elements is $\frac{5!}{2!2!1!} = 30$; ② Valid arrangement condition:

Adjacent colors are different, with the black ball (only 1) being the key to separating red and

white balls; ③ Count valid arrangements by black ball position:

- Black ball at position 2: Remaining positions 1,3,4,5 need to arrange 2 red and 2 white balls with no adjacent duplicates, only 1 arrangement (Red, Black, White, Red, White);
- Black ball at position 3: Remaining positions 1,2,4,5 need to arrange 2 red and 2 white balls with no adjacent duplicates, 2 arrangements (Red, White, Black, Red, White; White, Red, Black, White, Red);
- Black ball at position 4: Same as position 2, only 1 arrangement (Red, White, Red, Black, White);
- Black ball at position 1: Arrangement (Black, Red, White, Red, White), valid;
- Black ball at position 5: Arrangement (Red, White, Red, White, Black), valid; Valid arrangements: $1+2+1+1+1=6$;

④ Calculate probability: $P = \frac{6}{30} = \frac{1}{5}$;

3. Conclusion: So the answer is B.

易错点:1. 总样本数算为 $5! = 120$ (忽略球相同), 导致概率误算为 $\frac{6}{120} = \frac{1}{20}$; 2. 枚举有效排列时遗漏黑球在两端的情况, 或重复计数相同排列; 3. 未利用黑球的分隔作用, 直接枚举所有排列, 易出错。

1. Calculating the total number of sample points as $5! = 120$ (ignoring identical balls), leading

to a miscalculated probability of $\frac{6}{120} = \frac{1}{20}$; 2. Missing black ball at both ends when enumerating valid arrangements or double-counting identical arrangements; 3. Not using the separating effect of the black ball and directly enumerating all arrangements, which is error-prone.

第 17 题

题目内容:圆上 27 个点将圆等分, 任取 3 个不同的点, 顺次连接形成正三角形的概率为 ()

27 points on a circle divide the circle equally. Take 3 different points randomly and connect them in sequence. What is the probability of forming an equilateral triangle? ()

A. $\frac{1}{145}$ B. $\frac{1}{95}$ C. $\frac{1}{57}$ D. $\frac{1}{28}$

答案:B

详细解析:

1. 核心知识点: 古典概型、圆内接正三角形性质 (顶点需三等分圆周);

2. 解题步骤: ① 总样本数: 从 27 个点中选 3 个的组合数 $C_{27}^3 = \frac{27 \times 26 \times 25}{3 \times 2 \times 1} = 2925$; ② 正三角形计数: 圆 27 等分后, 相邻两点对应的圆心角为 $\frac{360^\circ}{27} = \frac{40^\circ}{3}$, 内接正三角形的顶点需满足“相邻顶点间隔 9 个点” ($27 \div 3 = 9$), 即每 9 个点取 1 个顶点; ③ 有效正三角形个数: 27 个点中, 每 3 个间隔 9 的点构成 1 个正三角形, 共 $27 \div 3 = 9$ 个 (避免重复计数); ④ 计算概率:

$$P = \frac{9}{2925} = \frac{1}{95};$$

3. 结论: 故**答案**为 B。

1. Core Knowledge: Classical probability model, property of regular inscribed triangles (vertices must trisect the circumference);

2. Problem-Solving Steps: ① Total number of sample points: The combination number of selecting 3 from 27 points is $C_{27}^3 = \frac{27 \times 26 \times 25}{3 \times 2 \times 1} = 2925$; ② Count of regular triangles: After 27

points equally divide the circle, the central angle between adjacent points is $\frac{360^\circ}{27} = \frac{40^\circ}{3}$. The vertices of a regular inscribed triangle must satisfy "adjacent vertices are 9 points apart" ($27 \div 3 = 9$), i.e., 1 vertex every 9 points; ③ Number of valid regular triangles: Among 27 points, every 3 points spaced 9 apart form 1 regular triangle, totaling $27 \div 3 = 9$ (avoid double-counting); ④ Calculate probability: $P = \frac{9}{2925} = \frac{1}{95}$;

3. Conclusion: So the answer is B.

易错点:1. 误将正三角形个数算为 27 (每个点为起点, 未除以 3 去重); 2. 不了解圆内接正三角形的顶点间隔规律, 直接枚举导致计数错误; 3. 组合数公式计算错误 (如 C_{27}^3 算为 $27 \times 26 \times 25$)。

1. Mistakenly counting the number of regular triangles as 27 (taking each point as the starting point without dividing by 3 to remove duplicates); 2. Not understanding the vertex spacing rule of regular inscribed triangles and directly enumerating leading to counting errors; 3. Miscalculating the combination formula (e.g., calculating C_{27}^3 as $27 \times 26 \times 25$).

第 18 题

题目内容:生物实验室有 5 只兔子, 其中 3 只测量过某项指标, 从 5 只中随机取出 3 只, 恰有 2 只测量过该指标的概率为 ()

A biology laboratory has 5 rabbits, 3 of which have been measured for a certain indicator. Take 3 rabbits randomly from the 5. What is the probability that exactly 2 of them have been measured for the indicator? ()

B. $\frac{2}{3}$ B. $\frac{3}{5}$ C. $\frac{2}{5}$ D. $\frac{1}{5}$

答案:B

详细解析：

1. 核心知识点：古典概型、组合计数（分层抽样）；

2. 解题步骤：① 总样本数：从 5 只兔子选 3 只的组合数 $C_5^3 = 10$ ；② 事件 A（恰有 2 只测量过）：从 3 只测量过的选 2 只，从 2 只未测量的选 1 只，组合数 $C_3^2 \times C_2^1 = 3 \times 2 = 6$ ；③

代入古典概型公式： $P(A) = \frac{\text{事件A样本数}}{\text{总样本数}} = \frac{6}{10} = \frac{3}{5}$ ；

3. 结论：故答案为 B。

1. Core Knowledge: Classical probability model, combination counting (stratified sampling);

2. Problem-Solving Steps: ① Total number of sample points: The combination number of selecting 3 from 5 rabbits is $C_5^3 = 10$ ；② Event A (exactly 2 measured): Select 2 from 3 measured rabbits and 1 from 2 unmeasured rabbits, combination number $C_3^2 \times C_2^1 = 3 \times 2 =$

6；③ Substitute into the classical probability formula: $P(A) = \frac{\text{Number of sample points in event A}}{\text{Total number of sample points}} = \frac{6}{10} = \frac{3}{5}$ ；

3. Conclusion: So the answer is B.

易错点:1. 混淆“恰有 2 只”和“至少 2 只”，误将 $C_3^2 \times C_2^1 + C_3^3$ （7 个）作为事件数，导致概率误算为 $\frac{7}{10}$ ；2. 组合数计算错误，如将 C_3^2 算为 2，导致事件数偏少；3. 用排列数计算样本数（如分母用 $5 \times 4 \times 3 = 60$ ），逻辑错误且结果偏差。

1. Confusing “exactly 2” with “at least 2”, mistakenly taking $C_3^2 \times C_2^1 + C_3^3$ (7) as the number of event points, leading to a miscalculated probability of $\frac{7}{10}$ ；2. Miscalculating combination numbers (e.g., $C_3^2 = 2$), resulting in fewer event points; 3. Using permutation numbers to calculate sample points (e.g., denominator $5 \times 4 \times 3 = 60$), which is logically incorrect and leads to deviations.

第 19 题

题目内容:盒中装有 3 个红球、2 个黄球, 随机取出 2 个球, 所取 2 个球颜色不同的概率等于

A box contains 3 red balls and 2 yellow balls. Take 2 balls randomly. What is the probability that the two balls are of different colors? ()

A. $\frac{3}{5}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{4}{5}$

答案:A

详细解析:

1. 核心知识点: 古典概型、对立事件 (简化计算);

2. 解题步骤: ① 总样本数: 从 5 个球中选 2 个的组合数 $C_5^2 = 10$; ② 定义对立事件: “颜色不同”的对立事件是“颜色相同”, 包含“全红”和“全黄”两类; ③ 计算对立事件样本数: 全红组合数 $C_3^2 = 3$, 全黄组合数 $C_2^2 = 1$, 对立事件总样本数 $3 + 1 = 4$; ④ 代入公式:

$$P(\text{颜色不同}) = 1 - P(\text{颜色相同}) = 1 - \frac{4}{10} = \frac{6}{10} = \frac{3}{5};$$

3. 结论: 故**答案**为 A。

1. Core Knowledge: Classical probability model, complementary events (simplified calculation);

2. Problem-Solving Steps: ① Total number of sample points: The combination number of selecting 2 from 5 balls is $C_5^2 = 10$; ② Define the complementary event: The complementary event of “different colors” is “same color”, including “all red” and “all yellow”; ③ Calculate the number of sample points for the complementary event: Combination number of all red is $C_3^2 = 3$, combination number of all yellow is $C_2^2 = 1$, total sample points for the complementary event $3 + 1 = 4$; ④ Substitute into the formula: $P(\text{Different colors}) = 1 - P(\text{Same color}) = 1 - \frac{4}{10} = \frac{6}{10} = \frac{3}{5}$; 3. Conclusion: So the answer is A.

易错点:1. 直接计算“1 红 1 黄”时, 误将组合数算为 $3 + 2 = 5$ (而非 $3 \times 2 = 6$), 导致概率

误算为 $\frac{5}{10} = \frac{1}{2}$; 2. 漏算全黄的组合数, 仅用全红样本数计算对立事件, 导致 $P(\text{颜色相同}) = \frac{3}{10}$, 进而误算目标概率为 $\frac{7}{10}$; 3. 混淆组合与排列, 用 $5 \times 4 = 20$ 作为总样本数, 结果错误。

1. When directly calculating "1 red and 1 yellow", mistakenly calculating the combination number as $3 + 2 = 5$ (instead of $3 \times 2 = 6$), leading to a miscalculated probability of $\frac{5}{10} = \frac{1}{2}$; 2. Missing the combination number of all yellow and only using the all-red sample points to calculate the complementary event, resulting in $P(\text{Same color}) = \frac{3}{10}$ and further miscalculating the target probability as $\frac{7}{10}$; 3. Confusing combination with permutation and using $5 \times 4 = 20$ as the total number of sample points, leading to incorrect results.

第 20 题

题目内容:三位同学参加跳高、跳远、铅球项目比赛, 每人选择其中两个项目, 则有且仅有两人选择的项目完全相同的概率是 ()

Three students participate in high jump, long jump, and shot put competitions. Each student chooses two of the events. What is the probability that exactly two students choose exactly the same two events? ()

D. $\frac{1}{3}$ B. $\frac{2}{3}$ C. $\frac{1}{2}$ D. $\frac{3}{4}$

答案:B

详细解析:

1. 核心知识点: 古典概型、分步计数原理;

2. 解题步骤: ① 总样本数: 每人从 3 个项目选 2 个, 有 $C_3^2 = 3$ 种选择 ({跳高,跳远}、{跳高,铅球}、{跳远,铅球}), 3 名同学的总选择数为 $3 \times 3 \times 3 = 27$; ② 事件 A (恰有两人项目相同): 分步计算:

- 第一步: 选相同的项目组合: 从 3 种组合中选 1 种, 有 $C_3^1 = 3$ 种;

- 第二步：选 2 名选择该组合的同学：从 3 名同学中选 2 名，有 $C_3^2 = 3$ 种；
- 第三步：剩余 1 名同学选不同组合：从剩余 2 种组合中选 1 种，有 2 种；
- 事件 A 样本数： $3 \times 3 \times 2 = 18$ ；

③ 代入公式： $P(A) = \frac{18}{27} = \frac{2}{3}$ ；

3. 结论：故答案为 B。

1. Core Knowledge: Classical probability model, step-by-step counting principle;

2. Problem-Solving Steps: ① Total number of sample points: Each student chooses 2 out of 3 events, with $C_3^2 = 3$ choices ({High Jump, Long Jump}, {High Jump, Shot Put}, {Long Jump, Shot Put}). The total number of choices for 3 students is $3 \times 3 \times 3 = 27$; ② Event A (exactly two students with the same event combination): Calculate step by step:

- Step 1: Select the same event combination: Choose 1 from 3 combinations, $C_3^1 = 3$ ways;
- Step 2: Select 2 students to choose this combination: Choose 2 from 3 students, $C_3^2 = 3$ ways;
- Step 3: The remaining 1 student chooses a different combination: Choose 1 from the remaining 2 combinations, 2 ways;
- Number of sample points for event A: $3 \times 3 \times 2 = 18$ ；

③ Substitute into the formula: $P(A) = \frac{18}{27} = \frac{2}{3}$ ；

3. Conclusion: So the answer is B.

易错点:1. 总样本数误算为 $P_3^2 \times 3 = 18$ （将选择视为有序），导致概率误算为 $\frac{18}{18} = 1$ ； 2. 事件数计算时漏乘“剩余同学的选择数”，仅算 $3 \times 3 = 9$ ，导致概率误算为 $\frac{9}{27} = \frac{1}{3}$ ； 3. 混淆“恰有两人相同”与“至少两人相同”，计入三人都相同的 3 种情况，事件数变为 21，概率误算为 $\frac{21}{27} = \frac{7}{9}$ 。

1. Miscalculating the total number of sample points as $P_3^2 \times 3 = 18$ (regarding choices as ordered), leading to a miscalculated probability of $\frac{18}{18} = 1$; 2. Missing the multiplication of “choices for the remaining student” when calculating event points, only calculating $3 \times 3 = 9$, leading to a miscalculated probability of $\frac{9}{27} = \frac{1}{3}$; 3. Confusing “exactly two students the same” with “at least two students the same”, including 3 cases where all three are the same, resulting in 21 event points and a miscalculated probability of $\frac{21}{27} = \frac{7}{9}$.

二、非选择题

II. Non-Multiple-Choice Questions

第 1 道非选择题

题目内容:判断下列解答是否正确, 并说明理由:

某运动员连续两次飞碟射击, 用 y 表示命中、 n 表示未命中, 样本空间 $\Omega = yy, yn, ny, nn$, 故“两次都命中”的概率为 0.25。

Judge whether the following solution is correct and explain the reason:

An athlete shoots at clay pigeons twice in a row. Let y represent a hit and n represent a miss.

The sample space is $\Omega = yy, yn, ny, nn$, so the probability of “hitting both times” is 0.25.

答案:不正确 Incorrect

详细解析:

1. 核心知识点: 古典概型的“等可能性”前提;
2. 解题步骤: ① 古典概型要求样本空间中每个样本点发生的概率相等; ② 该解答默认 yy, yn, ny, nn 四个样本点等可能, 但实际射击的“命中概率”(设为 p) 与“未命中概率”($1 - p$)不一定相等(如运动员命中率可能为 0.8, 未命中率为 0.2); ③ 此时 $P(yy) = p^2$, $P(yn) = P(ny) = p(1 - p)$, $P(nn) = (1 - p)^2$, 仅当 $p = 0.5$ 时 $P(yy) = 0.25$, 但题目未说明“命

中率为 0.5”;

3. 结论: 解答未验证样本点的等可能性, 故不正确。

1. Core Knowledge: The prerequisite of “equipotentiality” in classical probability model;

2. Problem-Solving Steps: ① The classical probability model requires that each sample point in the sample space has equal probability of occurrence; ② The solution assumes that the four sample points yy, yn, ny, nn are equally likely, but in reality, the “hit probability” (denoted as p) and “miss probability” ($1 - p$) of shooting are not necessarily equal (e.g., an athlete's hit rate may be 0.8 and miss rate 0.2); ③ At this time, $P(yy) = p^2$, $P(yn) = P(ny) = p(1 - p)$, $P(nn) = (1 - p)^2$, and $P(yy) = 0.25$ only when $p = 0.5$, but the question does not state that “the hit rate is 0.5”;

3. Conclusion: The solution fails to verify the equipotentiality of sample points, so it is incorrect.

易错点:忽略古典概型“等可能性”的核心前提, 误将样本空间的所有结果直接视为等可能, 未考虑实际射击中命中与未命中概率可能不相等的情况。

Ignoring the core prerequisite of “equipotentiality” in the classical probability model, mistakenly regarding all outcomes of the sample space as equally likely directly, without considering the possibility that the hit and miss probabilities may be unequal in actual shooting.

第 2 道非选择题

题目内容:从 52 张扑克牌 (不含大小王) 中随机抽 1 张, 计算下列事件的概率:

- (1) 抽到的牌是 7; (2) 抽到的牌不是 7; (3) 抽到的牌是方片; (4) 抽到 J 或 Q 或 K; (5) 抽到的牌既是红心又是草花; (6) 抽到的牌比 6 大比 9 小; (7) 抽到的牌是红花色; (8) 抽到的牌是红花色或黑花色。

Draw 1 card randomly from 52 poker cards (excluding jokers). Calculate the probability of the

following events:

- (1) Drawing a 7; (2) Not drawing a 7; (3) Drawing a diamond; (4) Drawing a J, Q, or K; (5) Drawing a card that is both a heart and a club; (6) Drawing a card larger than 6 and smaller than 9; (7) Drawing a red-colored card; (8) Drawing a red-colored card or a black-colored card.

答案: (1) $\frac{1}{13}$; (2) $\frac{12}{13}$; (3) $\frac{1}{4}$; (4) $\frac{3}{13}$; (5) 0; (6) $\frac{2}{13}$; (7) $\frac{1}{2}$; (8) 1

详细解析:

- 核心知识点: 古典概型、对立事件、互斥事件概率;
- 解题步骤 (总样本数 $n = 52$):
(1) 事件 A (抽到 7): 52 张牌中含 4 张 7 (红桃 7、方片 7、黑桃 7、草花 7), $n(A) = 4$, $P(A) = \frac{4}{52} = \frac{1}{13}$;
(2) 事件 B (不是 7): B 是 A 的对立事件, $P(B) = 1 - P(A) = 1 - \frac{1}{13} = \frac{12}{13}$;
(3) 事件 C (方片): 方片共 13 张, $n(C) = 13$, $P(C) = \frac{13}{52} = \frac{1}{4}$;
(4) 事件 D (J 或 Q 或 K): J、Q、K 各 4 张, 共 12 张 (互斥事件), $n(D) = 12$, $P(D) = \frac{12}{52} = \frac{3}{13}$;
(5) 事件 E (既是红心又是草花): 红心与草花无公共牌 (不可能事件), $n(E) = 0$, $P(E) = 0$;
(6) 事件 F (比 6 大比 9 小): 符合条件的牌是 7、8, 各 4 张共 8 张, $n(F) = 8$, $P(F) = \frac{8}{52} = \frac{2}{13}$;
(7) 事件 G (红花色): 红花色含红心、方片各 13 张, 共 26 张, $n(G) = 26$, $P(G) = \frac{26}{52} = \frac{1}{2}$;
(8) 事件 H (红花色或黑花色): 所有牌非红即黑 (必然事件), $n(H) = 52$, $P(H) = 1$;

3. 结论: 对应答案如上。

- Core Knowledge: Classical probability model, complementary events, probability of mutually exclusive events;
- Problem-Solving Steps (total sample size $n = 52$):
(1) Event A (Drawing a 7): There are 4 sevens in 52 cards (Heart 7, Diamond 7, Spade 7, Club 7), $n(A) = 4$, $P(A) = \frac{4}{52} = \frac{1}{13}$;
(2)

Event B (Not drawing a 7): B is the complementary event of A, $P(B) = 1 - P(A) = 1 - \frac{1}{13} = \frac{12}{13}$;

(3) Event C (Drawing a diamond): There are 13 diamonds, $n(C) = 13$, $P(C) = \frac{13}{52} = \frac{1}{4}$; (4)

Event D (Drawing a J, Q, or K): There are 4 Js, 4 Qs, and 4 Ks (mutually exclusive events), totaling

12 cards, $n(D) = 12$, $P(D) = \frac{12}{52} = \frac{3}{13}$; (5) Event E (Both heart and club): Hearts and clubs

have no common cards (impossible event), $n(E) = 0$, $P(E) = 0$; (6) Event F (Larger than 6

and smaller than 9): Qualified cards are 7 and 8, 4 each totaling 8 cards, $n(F) = 8$, $P(F) =$

$\frac{8}{52} = \frac{2}{13}$; (7) Event G (Red-colored card): Red cards include 13 hearts and 13 diamonds, totaling

26 cards, $n(G) = 26$, $P(G) = \frac{26}{52} = \frac{1}{2}$; (8) Event H (Red or black card): All cards are either red

or black (certain event), $n(H) = 52$, $P(H) = 1$;

3. Conclusion: The corresponding answers are as above.

易错点: 1. 计数错误 (如误将 J/Q/K 的数量算为 9 张); 2. 混淆“互斥事件”与“对立事件” (如事件(8)误算为 $\frac{1}{2} + \frac{1}{2} = 1$, 虽结果正确但逻辑不严谨, 应明确是必然事件); 3. 误将“既是红心又是草花”视为可能事件, 导致概率计算错误。

1. Counting errors (e.g., mistakenly calculating the number of J/Q/K as 9); 2. Confusing “mutually

exclusive events” with “complementary events” (e.g., miscalculating event (8) as $\frac{1}{2} + \frac{1}{2} = 1$,

which is correct in result but logically inaccurate; it should be clearly defined as a certain event);

3. Mistakenly regarding “both heart and club” as a possible event, leading to incorrect probability calculation.

第 3 道非选择题

题目内容:从 0~9 这 10 个数中随机选择一个数, 求下列事件的概率:

(1) 这个数平方的个位数字为 1; (2) 这个数的四次方的个位数字为 1。

Choose a number randomly from 0~9. Find the probability of the following events:

(1) The units digit of the square of this number is 1; (2) The units digit of the fourth power of this number is 1.

答案:(1) $\frac{1}{5}$; (2) $\frac{2}{5}$

详细解析:

1. 核心知识点: 古典概型、枚举法 (数字特征分析);

2. 解题步骤 (总样本数 $n = 10$): (1) 事件 A (平方个位为 1): 枚举 0~9 的平方个位: $0^2 = 0$ (0)、 $1^2 = 1$ (1)、 $2^2 = 4$ (4)、 $3^2 = 9$ (9)、 $4^2 = 16$ (6)、 $5^2 = 25$ (5)、 $6^2 = 36$ (6)、 $7^2 = 49$ (9)、 $8^2 = 64$ (4)、 $9^2 = 81$ (1); 符合条件的数为 1、9, 共 2 个,

$n(A) = 2$, $P(A) = \frac{2}{10} = \frac{1}{5}$; (2) 事件 B (四次方个位为 1): 四次方=平方的平方, 个位数字由平方个位决定: 平方个位为 1 $\rightarrow$ 四次方个位= $1^2=1$; 平方个位为 9 $\rightarrow$ 四次方个位= $9^2=81$ (个位 1); 其他平方个位 (0、4、5、6) 的四次方个位均不为 1; 符合条件的数为 1、3、7、9 ($3^2 = 9$ 、 $7^2 = 49$, 平方个位为 9, 四次方个位为 1), 共 4 个, $n(B) = 4$, $P(B) = \frac{4}{10} = \frac{2}{5}$;

3. 结论: 对应答案如上。

1. Core Knowledge: Classical probability model, enumeration method (digital characteristic analysis);

2. Problem-Solving Steps (total sample size $n = 10$): (1) Event A (Units digit of square is 1): Enumerate the units digits of squares of 0~9: $0^2 = 0$ (0), $1^2 = 1$ (1), $2^2 = 4$ (4), $3^2 = 9$ (9), $4^2 = 16$ (6), $5^2 = 25$ (5), $6^2 = 36$ (6), $7^2 = 49$ (9), $8^2 = 64$ (4), $9^2 = 81$ (1); Qualified numbers are 1 and 9, totaling 2, $n(A) = 2$, $P(A) = \frac{2}{10} = \frac{1}{5}$; (2) Event B (Units digit of fourth power is 1): Fourth power = square of square, and the units digit is determined by the square's units digit: If the square's units digit is 1 $\rightarrow$ fourth power's units digit = $1^2=1$; If the square's units

digit is 9 $\rightarrow$ fourth power's units digit = $9^2=81$ (units digit 1); The fourth power's units digit of other square units digits (0, 4, 5, 6) is not 1; Qualified numbers are 1, 3, 7, 9 ($3^2 = 9$, $7^2 = 49$, square units digit is 9, fourth power units digit is 1), totaling 4, $n(B) = 4$, $P(B) = \frac{4}{10} = \frac{2}{5}$;

3. Conclusion: The corresponding answers are as above.

易错点:1. 漏算四次方个位为 1 的数 (仅识别 1、9, 忽略 3、7); 2. 直接计算数字的四次方 (笨算), 导致计算复杂且易出错; 3. 混淆“平方个位”与“四次方个位”的逻辑关系, 未利用“四次方是平方的平方”简化分析。

1. Missing numbers with fourth power units digit 1 (only recognizing 1 and 9, ignoring 3 and 7);

2. Directly calculating the fourth power of numbers (brute calculation), leading to complex and

error-prone computation; 3. Confusing the logical relationship between “square units digit” and

“fourth power units digit”, failing to use “fourth power is the square of square” to simplify analysis.

第 4 道非选择题

题目内容:将一枚均匀的硬币投掷 6 次, 求正面出现的次数比反面出现的次数多的概率。

Toss a fair coin 6 times. Find the probability that the number of heads is greater than the number of tails.

答案: $\frac{11}{32}$

详细解析:

1. 核心知识点: 古典概型、二项分布、对称法;

2. 解题步骤: ① 总样本数: 每次投掷 2 种结果, 6 次共 $2^6 = 64$ 个等可能样本点; ② 定义事件: 设正面次数为 k , 反面次数为 $6 - k$, 目标事件为 $k > 6 - k$ (即 $k \geq 4$, k 为整数); ③ 利用对称法简化: - 事件 A ($k > 6 - k$) 与事件 B ($k < 6 - k$) 概率相等 (对

称), 设为 p ; - 事件 C ($k = 6 - k$) 即 $k = 3$ (正面=反面=3 次), 概率 $P(C) = \frac{C_6^3}{64} =$

$\frac{20}{64} = \frac{5}{16}$; - 由概率和为 1: $2p + P(C) = 1$, 解得 $p = \frac{1 - \frac{5}{16}}{2} = \frac{11}{32}$;

3. 结论: 正面次数比反面多的概率为 $\frac{11}{32}$ 。

1. Core Knowledge: Classical probability model, binomial distribution, symmetry method; 2.

Problem-Solving Steps: ① Total sample points: Each toss has 2 outcomes, 6 tosses have $2^6 =$

64 equally likely sample points; ② Define events: Let the number of heads be k and the

number of tails be $6 - k$. The target event is $k > 6 - k$ (i.e., $k \geq 4$, k is an integer); ③

Simplify using symmetry: - Event A ($k > 6 - k$) and Event B ($k < 6 - k$) have equal

probability (symmetry), denoted as p ; - Event C ($k = 6 - k$) means $k = 3$ (3 heads = 3

tails), probability $P(C) = \frac{C_6^3}{64} = \frac{20}{64} = \frac{5}{16}$; - Since the sum of probabilities is 1: $2p + P(C) = 1$,

solve for $p = \frac{1 - \frac{5}{16}}{2} = \frac{11}{32}$; 3. Conclusion: The probability that the number of heads is greater

than the number of tails is $\frac{11}{32}$ 。

易错点:1. 直接枚举 $k = 4, 5, 6$ 的样本点 (笨算), 导致组合数计算错误 (如 C_6^4 误算为 10);

2. 忽略对称法, 未利用“ A 与 B 概率相等”简化计算, 增加运算量; 3. 误将 $k \geq 3$ 视为目标

事件 (混淆“大于”与“大于等于”)。

1. Directly enumerating sample points for $k = 4, 5, 6$ (brute calculation), leading to errors in

combination number calculation (e.g., miscalculating C_6^4 as 10); 2. Ignoring the symmetry

method and failing to use “equal probability of A and B ” to simplify calculation, increasing the

computational load; 3. Mistaking $k \geq 3$ as the target event (confusing “greater than” with

“greater than or equal to”).

第 5 道非选择题

题目内容:在 1,2,...,2006 中随机选取三个数, 求能构成递增等差数列的概率。

Choose three numbers randomly from 1,2,...,2006. Find the probability that they can form an increasing arithmetic sequence.

答案: $\frac{3}{4010}$

详细解析:

1. 核心知识点: 古典概型、等差数列性质 ($2b = a + c$);

2. 解题步骤: ① 总样本数: 从 2006 个数中选 3 个, 组合数 $C_{2006}^3 = \frac{2006 \times 2005 \times 2004}{3 \times 2 \times 1}$; ② 有效样本数 (构成递增等差数列 $a < b < c$): - 由等差数列性质: $2b = a + c$, 故 a 与 c 同奇偶 ($a + c$ 为偶数); - 1~2006 中奇数、偶数各 1003 个 (2006 为偶数); - 选 2 个同奇偶的数 $a < c$, 则 $b = \frac{a+c}{2}$ 唯一确定, 且 b 在 1~2006 之间 (因 $a \geq 1$, $c \leq 2006$, 故 $b \geq \frac{1+3}{2} = 2$, $b \leq \frac{2004+2006}{2} = 2005$); - 有效样本数 = 奇数中选 2 个的组合数 + 偶数中选 2 个的组合数 = $2 \times C_{1003}^2 = 2 \times \frac{1003 \times 1002}{2} = 1003 \times 1002$; ③ 计算概率: $P = \frac{1003 \times 1002}{\frac{2006 \times 2005 \times 2004}{6}} = \frac{1003 \times 1002 \times 6}{2006 \times 2005 \times 2004}$; 化简 ($2006 = 2 \times 1003$, $2004 = 2 \times 1002$): $P = \frac{1003 \times 1002 \times 6}{2 \times 1003 \times 2005 \times 2 \times 1002} = \frac{6}{4 \times 2005} = \frac{3}{4010}$;

3. 结论: 所求概率为 $\frac{3}{4010}$ 。

1. Core Knowledge: Classical probability model, property of arithmetic sequence ($2b = a + c$);

2. Problem-Solving Steps: ① Total sample points: Combination number of selecting 3 from 2006 numbers, $C_{2006}^3 = \frac{2006 \times 2005 \times 2004}{3 \times 2 \times 1}$; ② Valid sample points (forming increasing arithmetic sequence $a < b < c$): - By the property of arithmetic sequence: $2b = a + c$, so a and c have the same parity ($a + c$ is even); - There are 1003 odd numbers and 1003 even numbers in 1~2006 (2006 is even); - Select 2 numbers $a < c$ with the same parity, then $b = \frac{a+c}{2}$ is uniquely determined and b is between 1~2006 (since $a \geq 1$, $c \leq 2006$, so $b \geq \frac{1+3}{2} = 2$, $b \leq \frac{2004+2006}{2} = 2005$); - Valid sample points = combination number of selecting 2 from odd

numbers + combination number of selecting 2 from even numbers = $2 \times C_{1003}^2 =$

$$2 \times \frac{1003 \times 1002}{2} = 1003 \times 1002 ; \textcircled{3} \text{ Calculate probability: } P = \frac{1003 \times 1002}{\frac{2006 \times 2005 \times 2004}{6}} = \frac{1003 \times 1002 \times 6}{2006 \times 2005 \times 2004} ;$$

$$\text{Simplify (} 2006 = 2 \times 1003 , 2004 = 2 \times 1002 \text{) : } P = \frac{1003 \times 1002 \times 6}{2 \times 1003 \times 2005 \times 2 \times 1002} = \frac{6}{4 \times 2005} = \frac{3}{4010} ;$$

3. Conclusion: The required probability is $\frac{3}{4010}$.

易错点:1. 未利用“ a 与 c 同奇偶”的性质, 直接枚举等差数列, 导致计数错误; 2. 组合数化简时未约分 (如未发现 $2006 = 2 \times 1003$), 导致计算复杂; 3. 误将“递增等差数列”视为“任意等差数列”, 重复计数 (如 a, b, c 与 c, b, a)。

1. Failing to use the property that “ a and c have the same parity” and directly enumerating arithmetic sequences, leading to counting errors;
2. Not canceling fractions when simplifying combination numbers (e.g., not noticing $2006 = 2 \times 1003$), resulting in complex calculations;
3. Mistaking “increasing arithmetic sequence” for “arbitrary arithmetic sequence” and double-counting (e.g., a, b, c and c, b, a).